

FROTIN[®]

FILM COATED TABLET 200MG

Ingredient(s):

Each film coated tablet contains:

Metronidazole 200mg

Pharmacodynamics:

1. Metronidazole is an antimicrobial drug that is primarily active against obligate anaerobic microorganisms, both bacteria and protozoa. It interacts with DNA to cause a loss of helical structure, strand breakage, and resultant inhibition of nucleic acid synthesis and cell death.
2. Metronidazole is very active against the clinically important protozoa *Trichomonas vaginalis*, *Giardia lamblia* and *Entamoeba histolytica*. It is also active against strains of *Bacteroides fragilis* and related species, *Fusobacterium*, *Clostridium*, *Peptococcus*, and *Peptostreptococcus*.

Pharmacokinetics:

1. Metronidazole is rapidly and almost completely absorbed on administration; peak plasma concentrations occur after 20 min to 3 hrs.
2. Oral administration of 200mg produced peak plasma concentrations of 4mcg/mL. The elimination half-life of metronidazole is 8.5 ± 2.9 hrs.
3. Metronidazole is widely and rapidly distributed in the body. It diffuses to the bile, breast milk, synovial fluid and saliva.
4. Metronidazole is metabolized primarily in the liver to various oxidative metabolites which are excreted partially conjugated as glucuronides in the urine. The major metabolites of metronidazole are 1-(2-hydroxyethyl)-2-hydroxymethyl-5-nitroimidazole, the hydroxy metabolite, and 1-acetic acid-2-methyl-5-nitroimidazole.
5. The major route of elimination of metronidazole and its metabolites is via the urine (60 to 80% of the dose), with faecal excretion accounting for 6 to 15% of the dose. Less than 20% of the circulating metronidazole is bound to plasma proteins.
6. Metronidazole can be used in chronic renal failure; it is rapidly removed from the plasma by dialysis.
7. Metronidazole is excreted in milk but the intake of a suckling infant of a mother receiving normal dosage would be considerably less than the therapeutic dose for infants.

Indication(s):

For the treatment of urogenital trichomoniasis in the female (*Trichomonas vaginalis*) and in the male. All forms of amoebiasis (intestinal and extraintestinal disease and that of symptomless cyst passers), acute ulcerative gingivitis and acute dental infection.

For the treatment and prophylaxis of anaerobic bacterial infection.

Contraindications:

Metronidazole is contraindicated in patients with a prior history of hypersensitivity to metronidazole or other nitroimidazole derivatives.

Dosage and Administration:

Metronidazole tablets should be swallowed whole (not chewed) with plenty of water. It is recommended that the tablets be taken during or after a meal. Other antibiotics may be used concurrently.

Adults:

a. *Trichomonas vaginitis*:

2 tablets twice daily for 7 days, or

4 tablets in the morning and 6 tablets in the evening for 2 days or

10 tablets as a single dose.

b. *Amoebiasis*:

• *Invasive intestinal disease in susceptible subject*:

Adult & children > 10 yr : 4 tablets thrice daily.

Children 7 – 10 yr : 2 tablets thrice daily.

Children 3 – 7 yr : 1 tablet four times daily.

Children 1-3 yr : 1 tablet thrice daily.

Duration of treatment: 5 days.

• *Intestinal disease in less susceptible subjects and chronic amoebic hepatitis*:

Adult & children > 10 yr : 2 tablets thrice daily.

Children 7-10 yr: 1 tablet thrice daily.

Children 3-7 yr : ½ tablet four times daily.

Children 1-3 yr : ½ tablet thrice daily.

Duration of treatment: 5-10 days.

• *Amoebic liver abscess, also other forms of extra intestinal amoebiasis*:

Adult & Children > 10 yr : 2 tablets thrice daily.

Children 7-10 yr: 1 tablet thrice daily.

Children 3-7 yr : ½ tablet four times daily.

Children 1-3 yr : ½ tablet thrice daily.

Duration of treatment: 5 days.

• *Symptomless cyst passer*:

Adults & Children > 10 yr : 2 - 4 tablets thrice daily.

Children 7-10 yr: 1-2 tablets thrice daily.

Children 3-7 yr : ½ -1 tablet four times daily.

Children 1-3 yr : ½ -1 tablet thrice daily.

Duration of treatment: 5 - 10 days.

c. *Acute ulcerative gingivitis*:

Adult & Children > 10 yr: 1 tablet thrice daily.

Children 3-10 yr: ½ tablet thrice daily.

Duration of treatment: 3 days.

d. *Acute dental infection*:

Adult & Children > 10 yr : 1 tablet thrice daily.

Duration of treatment: 3-7 days.

e. *Anaerobic bacterial infection*:

The duration of a course of metronidazole treatment is about 7 days but it will depend upon the seriousness of the patient's condition as assessed clinically and bacteriologically.

• *Prophylaxis against anaerobic infection, chiefly in the context of abdominal (especially colorectal) and gynaecological surgery*:

Adults: 2 tablets thrice daily for 3-4 days followed by postoperative IV or rectal administration until the patient is able to take tablets.

Children: 7.5mg/kg thrice daily.

• *Treatment of established anaerobic infection*:

Adults: 4 tablets initially followed by 2 tablets thrice daily.

Children: 7.5mg/kg thrice daily.

Route of Administration:

Oral.

Warnings and Precautions:

1. Caution should be observed in patients with severe hepatic disease since they metabolize metronidazole slowly which may result in accumulation of metronidazole and its metabolites in the plasma.
2. Cases of severe hepatotoxicity/acute hepatic failure, including cases with a fatal outcome with very rapid onset after treatment initiation in patients with Cockayne syndrome have been reported with products containing metronidazole for systemic use. In this population, metronidazole should therefore be used after careful benefit-risk assessment and only if no alternative treatment is available. Liver function tests must be performed just prior to the start of therapy, throughout and after end of treatment until liver function is within normal ranges, or until the baseline values are reached. If the liver function tests become markedly elevated during treatment, the drug should be discontinued. Patients with Cockayne syndrome should be advised to immediately report any symptoms of potential liver injury to their physician and stop taking metronidazole.
3. Metronidazole is a nitroimidazole and should be used with care in patients with evidence of or history of blood dyscrasia.
4. Safe use of this product during pregnancy and lactation has not been established. It should be given to women who suspect pregnancy or to pregnant women and nursing mothers only when clearly needed.
5. Regular clinical and laboratory monitoring are advised if administration of metronidazole for > 10 days is considered to be necessary.
6. There is a possibility that after *Trophomonas vaginalis* has been eliminated, a gonococcal infection might persist.
7. Clinicians who contemplate continuous therapy for the relief of chronic conditions for periods longer than those recommended are advised to consider the possible therapeutic benefit against the risk of peripheral neuropathy.
8. No routine adjustments in dosage of metronidazole need to be made in patients with renal failure undergoing intermittent peritoneal dialysis (IPD) or continuous ambulatory peritoneal dialysis (CAPD).
9. Caution is required in patients with active disease of the central nervous system except for brain abscess.

Side Effect(s) / Adverse Reaction(s):**Gastrointestinal system:**

- Gastrointestinal upset, including anorexia, nausea, vomiting and epigastric pain.

Mouth:

- A sharp, unpleasant metallic taste. Furry tongue, glossitis, and stomatitis may occur.

Hematopoietic:

- Reversible neutropenia (leucopenia) has been reported.

Central Nervous System:

- Convulsive seizures, peripheral neuropathy, dizziness, ataxia, confusion, irritability, depression, weakness, and insomnia may occasionally occur.

Hypersensitivity:

- Urticaria, erythematous rash, flushing, nasal congestion, dryness of mouth (or vagina or vulva), and fever.

Others:

Proliferation of *Candida* in the vagina, dyspareunia, decrease of libido, proctitis and fleeting joint pains may occur rarely.

Effects on Ability to Drive and Use Machines:

Patients should be warned about the potential for drowsiness, dizziness, confusion, hallucinations, convulsions or transient visual disorders, and advised not to drive or operate machinery if these symptoms occur.

Pregnancy and Lactation:

There is inadequate evidence of the safety of metronidazole in pregnancy but it has been in wide use for many years without apparent ill consequence. Nevertheless metronidazole, like other medicines, should not be given during pregnancy or during lactation unless the physician considers it essential; in these circumstances the short, high-dosage regimens are not recommended.

Drug Interactions:

1. Alcohol beverages should not be consumed during metronidazole therapy and for at least one day after because it may cause disulfiram-like reactions.
2. Metronidazole potentiates anticoagulant effect of warfarin-type oral anticoagulants, resulting in a prolongation of prothrombin time.
3. Phenobarbital increases the metabolism of metronidazole and resulting in a decrease in plasma concentration.
4. Plasma levels of lithium may be increased.

Symptoms and Treatment for Overdosage, and Antidote(s):

Symptoms of metronidazole overdose include nausea, vomiting, and ataxia. Neurotoxic effects, including seizures and peripheral neuropathy, have been reported after 5 to 7 days of doses of 6 to 10.4 g every other day. Uneventful recovery has followed attempts at suicide with quantities of 30 and 60 x 200mg tablets.

There is no specific antidote for its overdose. Early gastric lavage may remove a large amount of the drug; otherwise; symptomatic and supportive therapy should be initiated.

Shelf-Life:

3 years from the date of manufacture.

Storage Condition(s):

Store at temperature below 30°C. Protect from light and moisture.

Product Description:

Frothin Film Coated Tablet 200mg:

A white color round film coated tablet, one side impressed with "YSP".

Packing(s):

Blister packing of 10's x 10 and 10's x 100.



Manufacturer and Product Registration Holder:
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